

1. Derived categories.

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2. Exceptional collections and mutations.

- 1). Definitions of Exceptional collections and mutations
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② Serre functors..

3). An example: $D^b(\mathbb{C}P^n)$.

3. Euler pairing.

1). $K_0(X) = K_0(D^b(\text{Coh } X)) = K_0(\text{Coh } X)$ Grothendieck groups

2). Exceptional ~~sets~~ collections. and Grothendieck groups.

$$\{E_1, \dots, E_n\} \xrightarrow{\text{a basis for}} K_0(\text{Coh } X) \cong \mathbb{Z}^n.$$

3). Mutations $\rightarrow$ "reflections".

1. Triangulated categories and derived categories.

• Triangulated categories.

Def A triangulated category $\mathcal{D}$ is an additive category with an automorphism equivalence T functor (shift) and a set of exact (distinguished) triangles.

$$A \rightarrow B \rightarrow C \rightarrow T[A]$$

which subject to axioms TR1-TR4.

only mention TR1. TR3.

Part of TR1: Any morphism $A \xrightarrow{u} B$ can be completed to be an exact triangle. i.e.

$$A \xrightarrow{u} B \rightarrow C \rightarrow A[1]$$

TR3. Suppose we have f, g such that there exists a commutative diagram.

$$\begin{array}{ccccccc} A & \rightarrow & B & \rightarrow & C & \rightarrow & A[1] \\ f \downarrow & & g \downarrow & & h \downarrow & & \downarrow f[1] \\ A' & \rightarrow & B' & \rightarrow & C' & \rightarrow & A'[1] \end{array}$$

then ~~there exists a morphism~~ there exists a morphism $h: C \rightarrow C'$ s.t. diagram commutes.

Exact functors between triangulated categories.

$F: \mathcal{D} \rightarrow \mathcal{D}'$ s.t. 1) $F \circ T_{\mathcal{D}} \simeq T_{\mathcal{D}'} \circ F$ 2) F maps exact triangles to exact triangles.

• Derived categories. and ~~abelian categories~~

Denote $\text{Kom}(\mathcal{A})$ the ~~chain~~ category of chain complexes in $\mathcal{A}$ (abelian category). There is a natural shift functor on $\text{Kom}(\mathcal{A})$.

Def / Theorem derived categories. There exists a category $\mathcal{D}(\mathcal{A})$ and a functor.

$Q: \text{Kom}(\mathcal{A}) \rightarrow \mathcal{D}(\mathcal{A})$ s.t.

i). If $f: A' \rightarrow B'$ is a quasi isomorphism, then $Q(f)$ is an isomorphism in $\mathcal{D}(\mathcal{A})$.

ii). Any functor satisfying i). factorizes uniquely over Q .

$$\begin{array}{ccc} \text{Kom}(\mathcal{A}) & \xrightarrow{Q} & \mathcal{D}(\mathcal{A}) \\ & \searrow F & \uparrow \exists! G \\ & & \mathcal{D} \end{array}$$

$$\text{Kom}(\mathcal{A}) \xrightarrow{\text{chain homotopy equivalence}} K(\mathcal{A}) \xrightarrow{\text{localisation}} \mathcal{D}(\mathcal{A}) = S^{-1}K(\mathcal{A})$$

S is the collection of all quasi isomorphism.
triangulated $\mathcal{D}(\mathcal{A})$ is also triangulated.

- Example $D^b(X) = D^b(\text{coh } X)$ $D^b(\emptyset) = \langle 0, \dots, 0(n) \rangle$ $D^b(\mathbb{CP}^1) = \langle 0, \mathcal{O}(1) \rangle$
 mention one point: $D^+(Q\text{coh}(X)) \xrightarrow{\uparrow} D^b(\text{coh}(X)) \xrightarrow{\downarrow} D^L(\text{coh}(X))$
 has enough injectives. $\downarrow$ allows us inclusion equivalence
 $\text{Hom}_{D^b(X)}(A, B) \simeq \text{Ext}_{D^b(X)}^i(A, B)$

For $\mathbb{CP}^n$
 $\text{Ext}^i(\mathcal{O}, \mathcal{O}(m)) = \begin{cases} 0 & i \neq 0, n \\ \binom{n+m}{n} & i = 0, m \geq 0 \\ \binom{n}{-m} & i = n, m \leq 0 \end{cases}$
 Serre duality.

$\mathbb{CP}^2$	-3	-2	-1	$m=0$	1	2
				1.	3	6.
			1			
	6	3	1			

derived functors

X smooth variety. $\Gamma: Q\text{coh}(X) \rightarrow \text{Vect}(k)$ $\Gamma \rightarrow \Gamma(X, F)$ global section. left exact

Cohomology

take define right derived ~~category~~ functor

$RP: D^+(Q\text{coh}(X)) \rightarrow D^+(\text{Vect}(k))$

$R^i P(F) := H^i(X, F)$ hypercohomology groups

$RP: D^b(X) \xrightarrow{\text{inclusion}} D^b(Q\text{coh}(X)) \xrightarrow{RP} D^b(\text{Vect}(k))$

$D^+(Q\text{coh}(X)) \xrightarrow{RP} D^+(\text{Vect}(k))$

because of F is ~~not~~ quasi coherent sheaf then $H^i(X, F) = 0 \quad i > \dim(X)$

$\Rightarrow Rf_* \quad Lf^* \quad \otimes^L$
 right derived $\uparrow$ left derived

derived functors

Fourier-Mukai Transform

$Rf_* (R^i f^* L) \simeq R^i f_* (R^i f^* L)$

2. Exceptional collections and mutations.

2.1 Notations K is a field.

- $D^b(K)$: derived category of finite dimensional K -vector space. $\simeq \text{GrVect}_K^{\infty}$: category of finite dimensional $\mathbb{Z}$ -graded vector space. by.

$$\Phi: \text{GrVect}_K^{\infty} \rightarrow D^b(K). \quad V^* \mapsto \bigoplus_{i \in \mathbb{Z}} \frac{(\text{Gr}^i V^*)[-i]}{V[i]k}. \quad \text{Gr}^i V[-k] := \text{Gr}^{i+k} V$$

$$H^*: D^b(K) \rightarrow \text{GrVect}_K^{\infty} \quad F^* \mapsto H^*(F^*).$$

- $\mathcal{D}$: K -linear category of finite type: $\text{Hom}^*(X, Y) := \bigoplus_{i \in \mathbb{Z}} \text{Hom}^{\oplus}(X, Y[i])$. is finite dimensional K -linear space. for all X, Y in $\mathcal{D}$

$$V^* \otimes X := \bigoplus_i V^i \otimes X[-i]. \quad V \otimes X[-i] = \underbrace{X[-i] \otimes \dots \otimes X[-i]}_{\dim_K V \text{ times}}.$$

$$\text{S.t.} \quad \Downarrow \quad \boxed{\text{Hom}^*(Y, V^* \otimes X) = V^* \otimes \text{Hom}^*(Y, X). \quad \forall Y \in \text{Ob}(\mathcal{D})}$$

$$V^*[j] \otimes X[k] = (V^* \otimes X)[j+k]. \quad (V^*[j])^\vee = (V^*)^\vee[j].$$

- linear exact functor between two K -linear categories: $F: \mathcal{D} \rightarrow \mathcal{E}$.

$$\text{covariant} \quad F(V^* \otimes X) = V^* \otimes F(X). \quad \forall V^*, X \in \mathcal{D}.$$

$$\text{contravariant} \quad F(V^* \otimes X) = (V^*)^\vee \otimes F(X). \quad (F: \mathcal{D}^{\text{op}} \rightarrow \mathcal{E}).$$

$$b: \text{functor } \text{Hom}(-, -) : \mathcal{D} \times \mathcal{D}^{\text{op}} \rightarrow \text{GrVect}_K^{\infty} \quad \text{Hom}^*(W^* \otimes X, V^* \otimes Y) = (W^*)^\vee \otimes X^* \otimes \text{Hom}^*(X, Y).$$

$$j^*(X, Y) : \text{Hom}^*(X, Y) \otimes X \rightarrow Y. \quad j_*(X, Y) : \text{Hom}^*(X, Y)^\vee \otimes Y.$$

$$(\text{End}(\text{Hom}^*(X, Y))) = \text{Hom}^*(\text{Hom}^*(X, Y) \otimes X, Y) = \text{Hom}(X, \text{Hom}^*(X, Y)^\vee \otimes Y)$$

$$\text{prop} \quad \forall E \in \text{Ob}(\mathcal{D}). \quad \Phi_E: D^b(K) \rightarrow \mathcal{D} \quad \Phi_E^\vee: \mathcal{D} \rightarrow D^b(K). \quad \Phi_E!: \mathcal{D} \rightarrow L^b(K) \\ V^* \mapsto V^* \otimes E \quad X \mapsto \text{Hom}(X, E)^\vee \quad X \mapsto \text{Hom}(E$$

$$\text{adjunctions.} \quad \Phi_E^\vee \dashv \Phi_E \dashv \Phi_E!.$$

- Generated. Triangulated subcategory: $\Omega \subseteq \text{Ob}(\mathcal{D})$. $\langle \Omega \rangle$ the smallest full triangulated subcategory of $\mathcal{D}$ containing all objects of Ω .

$$\langle \Omega \rangle = \bigcup_{r \in \mathbb{N}^*} \Omega_r. \quad \Omega_1 := \{ X[n] \mid X \in \Omega, n \in \mathbb{Z} \}.$$

$$\Omega_r := \underbrace{\Omega_1 * \dots * \Omega_1}_{r \text{ times}}.$$

$$\Omega * \mathcal{B} = \{ X \mid A \rightarrow X \rightarrow B \rightarrow A[1] \text{ for some } A \text{ and } B \in \mathcal{B} \}. \\ A, B \in \text{Ob}(\mathcal{D})$$

2.2. Exceptional collections.

- Def. (Exceptional objects). $E \in \text{Ob}(\mathcal{D})$ is exceptional, if $\text{Hom}^i(E, E) = \begin{cases} \mathbb{K} & i=0 \\ 0 & i \neq 0 \end{cases}$.

Or more specifically $\text{Hom}(E, E[l]) = \begin{cases} \mathbb{K} & l=0 \\ 0 & l \neq 0 \end{cases}$.

$\{E_1, E_2, \dots, E_n\}$ is an exceptional collection, if $\text{Hom}^i(E_j, E_i) = 0$ whenever $i < j$.

more specifically. $\text{Hom}(E_j, E_i[l]) = \begin{cases} \mathbb{K} & \text{if } l=0 \text{ and } i=j \\ 0 & \text{if } i < j \text{ or if } l \neq 0, i=j \end{cases}$.

An exceptional collection is full if $\mathcal{D} = \langle E_1, E_2, \dots, E_n \rangle$. ($\langle E_1, E_2, \dots, E_n \rangle \rightarrow \mathcal{D}$ is equivalent, (of length n).

- Remark. $\text{Aut}(\mathcal{D})$ (and $\mathbb{Z}^n$) acts on the set of exceptional collection.
- and preserves fullness.

- Braid group B_n ^(left) acts on the set of exceptional collections of length n.

2.3. Mutations.

let E be an exceptional object. For any $X \in \text{Ob}(\mathcal{D})$.

define $L_E X := \text{cone}(\bigoplus_{i \in \mathbb{Z}} E^i(X) \rightarrow X)$ $R_E X := \text{cone}(X \rightarrow \bigoplus_{i \in \mathbb{Z}} E^i(X))$

In fact. we have exact triangles.

$$L_E X[-1] \rightarrow \text{Hom}(E, X) \otimes E \xrightarrow{j^*} X \rightarrow L_E X.$$

$$X \xrightarrow{j^*} \text{Hom}(X, E)^\vee \otimes E \xrightarrow{\text{apply Hom}(-, E)} L_E X$$

(By TR1 for triangulated category).

Facts. 1) $\text{Hom}(E, L_E X) = 0$ $\text{Hom}(R_E X, E) = 0$

2) $L_E X$ and $R_E X$ is unique up to unique isomorphism

$$\begin{array}{ccccccc} \text{Hom}(E, X) \otimes E & \xrightarrow{j^*} & X & \rightarrow & S & \rightarrow & (\text{Hom}(E, X) \otimes E)[1] \\ \downarrow & & \downarrow & & \downarrow f \leftarrow \text{unique isomorphism} & & \downarrow \\ \text{Hom}(E, X) \otimes E & \xrightarrow{j^*} & X & \rightarrow & L_E X & \rightarrow & (\text{Hom}(E, X) \otimes E)[1] \end{array}$$

apply $\text{Hom}(-, L_E X)$. (last term is zero by $\text{Hom}(E, L_E X) = 0$).

3) if $\text{Hom}(E, X) = 0$ ($\text{Hom}(X, E) = 0$), then $L_E X = X$ ($R_E X = X$)

$\downarrow$ $\downarrow$
 $X \in E^\perp$ $X \in {}^\perp E$

Prop (Braid group action). The mutations preserve ~~exceptionally~~ exceptionality.
and satisfy. $L_i R_i = R_i L_i = \text{Id.} (1).$

$$L_i L_j = L_j L_i \text{ for } |i-j| > 1 \quad (2).$$

$$L_{i+1} L_i L_{i+1} = L_i L_{i+1} L_i \text{ for } 1 < i < n. \quad (3).$$

$\Rightarrow$ B_{n+1} acts on the set of exceptional collections of length $(n+1)$:
 $(E_0, \dots, E_n)$ exceptional collection

$$L_i (E_0, \dots, E_n) := (E_0, \dots, L_{E_{i-1}} E_i, E_i, E_{i+1}, \dots, E_n).$$

$$R_i (E_0, \dots, E_n) := (E_0, \dots, E_i, R_{E_i} E_{i-1}, \dots, E_n).$$

Sketch of proof. For (1) (2) we only need to prove for a pair. (E_1, E_2) :

$$\begin{aligned} \text{Hom}^*(E, L_E X) = 0 &\leadsto \text{Hom}^*(E_1, L_{E_1} E_2) = 0 \\ \text{Hom}^*(R_{E_2} E_1, E_2) = 0 &\leadsto \text{Hom}^*(R_{E_2} X, E) = 0 \end{aligned} \quad \left. \vphantom{\begin{aligned} \text{Hom}^*(E, L_E X) = 0 \\ \text{Hom}^*(R_{E_2} E_1, E_2) = 0 \end{aligned}} \right\} \Rightarrow \text{preserve exceptionality}$$

$$\begin{aligned} R \circ L (E_1, E_2) &= (E_1, E_2). \quad \boxed{R_{E_1} E_2 = E_2} \text{ since } \text{Hom}(E_2, E_1) = 0. \\ \downarrow & \\ R(L_{E_1} E_2, E_1) &\rightarrow (E_1, R_{E_1}(L_{E_1} E_2)) \quad L_{E_2} E_1 = E_1 \text{ since } \text{Hom}(E_2, E_1) = 0. \end{aligned}$$

apply a fact $\text{Hom}^*(L_{E_1} X[-1], E)^* = \text{Hom}^*(E, R_{E_1} X)$.

$$\text{Hom}^*(L_{E_1} E_2[-1], E_1)^* = \text{Hom}^*(E_1, R_{E_1} E_2) = \text{Hom}^*(E_1, E_2).$$

first say

$$\begin{array}{ccccccc} L_{E_1} E_2[-1] & \rightarrow & \text{Hom}^*(E_1, E_2) \otimes E_1 & \rightarrow & E_2 & \rightarrow & L_{E_1} E_2 \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ L_{E_1} E_2[-1] & \rightarrow & \text{Hom}^*(L_{E_1} E_2[-1], E_1) & \rightarrow & (R_{E_1} L_{E_1} E_2) & \rightarrow & L_{E_1} E_2 \end{array}$$

$$\Rightarrow R \circ L (E_1, E_2) = (E_1, E_2). \quad L \circ R (E_1, E_2) = (E_1, E_2).$$

(2) is trivial. $\begin{array}{c} L_{i+1} L_{j+1} \\ \text{---} (E_i, E_{i+1}, \dots, E_j, E_{j+1}, \dots) \\ \downarrow L_{E_i} \quad \downarrow L_{E_j} \\ L_{E_i} E_{i+1}, E_i \quad L_{E_j} E_{j+1}, E_j \end{array}$

(3). $L_{i+1} L_i L_{i+1} = L_i L_{i+1} L_i$ $1 < i < n$.

$$\begin{array}{l} \xrightarrow{L_1} (L_{AB}, A, C) \xrightarrow{L_2} ((L_{AB}, L_{AC}, A) \xrightarrow{L_1} (L_{A(L_{AB}L_{AC})}, L_{AB}, A) \\ (A, B, C) \searrow \xrightarrow{L_2} (A, L_{BC}, B) \xrightarrow{L_1} (L_{A(L_{BC})}, A, B) \xrightarrow{L_2} (L_{A(L_{BC})}, L_{AB}, A) \end{array}$$

$$\boxed{L_{(AB)}(L_C)} = L_A(L_{BC}).$$

$$(L_E|_{\perp E} : \perp E \rightarrow E^\perp \text{ exact functor})$$

$$\boxed{L_A L_B L_C[-1]} \rightarrow \text{Hom}^*(L_A B, L_A C) \otimes L_A B. \rightarrow L_A C. \rightarrow \boxed{L_A L_B C}.$$

↑ apply $(L_A | L_A)$ and note that

$$L_B[-1] \rightarrow \text{Hom}(B, C) \otimes B \rightarrow C \rightarrow L_B C \quad (\text{triangle for } L_B C)$$

Braid group action: $\beta_{i,j}$ generators of B_{n+1} $1 \leq i \leq n$. $\beta_{i,j}$ exact linear functor #.

Let's $\beta_{i,i+2} \beta_{i,i+1} = \beta_{i,i+1} \beta_{i,i+2}$ $|i-j| \geq 1$ for a exceptional collection $(E_0, \dots, E_n)$ we have

relations $\beta_{i,i+2} \beta_{i,i+1} \beta_{i,i+2} = \beta_{i,i+1} \beta_{i,i+2} \beta_{i,i+1}$ $(E_0, \dots, E_n) \beta_{i,i+2} = L_i(E_0, \dots, E_n)$

$$\underline{\text{Lth}} \quad \beta_{i, i+2} \beta_{j, j+1} = \beta_{j+1, j} \beta_{i+1, i} \quad 1 \leq i \leq n, \quad |i-j| \geq 1$$

relations -

$$\beta_{i,i+2} \beta_{i+1,i+2} \beta_{i,i+1} = \beta_{i+1,i+2} \beta_{i,i+1} \beta_{i,i+2}$$

- Serre functor. Def. A Serre-functor in a $\mathbb{K}$ -linear category of finite.

type $\mathcal{D}$ is a $\mathbb{K}$ -linear ~~automorphism~~ auto equivalence $\mathcal{S} = \mathcal{D} \rightarrow \mathcal{D}$ s.t.

$$\text{Hom}(A, B) \xrightarrow{\cong} \text{Hom}(B, SA)^V \quad \forall A, B.$$

$\cong$ is functorial in A, B .

Thm Prop. Any Serre functor on a triangulated category $\mathcal{D}$ is exact, $\dots$ over a field k .

① necessary and sufficient condition for existence of Serre functor.

all functors. $\text{Hom}(X, -)^V$ and $\text{Hom}(-, X)^V$ are representable.

for any object $X \in \text{ob}(\mathcal{B})$.

② Any two same functor are connected by a canonical functor isomorphism

- In the case ~~of~~ ^{that}, D has a full exceptional collection, it admits

a Serre functor,

projective.

- If X is a smooth variety, then there exists a Serre functor on the $\mathcal{L}^b(X) = \mathcal{L}^b(\text{Coh } X)$ defined by

$S: D^b(X) \xrightarrow[\otimes \omega_X]{\omega_X \otimes -} D^b(X) \xrightarrow{[n]} D^b(X)$ where ω_X is the canonical bundle and $[n]$ is the shift by ~~half~~ dimension of X .

ω_X is a vector bundle thus locally free sheaf $\Rightarrow \otimes \omega_X$ is exact on $\text{coh}(X)$

Prop $(E_0, \dots, E_n)$ is a ^{full} exceptional collection then $(SE_n, E_0, \dots, E_{n-1})$ is

also a exceptional collection and $SE_n = L_{E_0} \dots L_{E_{n-1}} E_n$

Proof $\text{Hom}^i(SE_n, E_j) = \text{Hom}(E_j, SE_n[-i]) = \text{Hom}(\bigoplus E_n[-i], E_j)^\vee$

Since $n \neq j \Rightarrow \text{Hom}(E_n[-i], E_j) = 0$. $(\text{Hom}(SE_n, SE_n) \xrightarrow{\vee} \text{Hom}(E_n, SE_n)^\vee = \text{Hom}(E_n, E_n)^\vee)$

$\text{Hom}^i(E_n, X) = \text{Hom}^i(L_{E_0} \dots L_{E_{n-1}} E_n, X)^\vee = \text{Hom}(X, L_{E_0} \dots L_{E_{n-1}} E_n)^\vee$

$\text{Hom}^i(X, SE_n)^\vee$ by Yoneda. Lemma only need to check $\text{Hom}(E_n, X) \otimes E_n \rightarrow X \rightarrow L_{E_n} X$ $(\text{Hom}(X, SE_n) = \text{Hom}(X, L_{E_0} \dots L_{E_{n-1}} E_n)^\vee)$

$\text{Hom}^i(X, L_{E_0} \dots L_{E_{n-1}} E_n) \cong \text{Hom}^i(E_n, X)^\vee \otimes \text{Hom}^i(E_n, L_{E_0} \dots L_{E_{n-1}} E_n)$

note that $\text{Hom}(E'_0, L_{E_n} X) = \text{Hom}(E'_0, X)$ if $E'_0 \perp E_n$
 $\Rightarrow \text{Hom}^i(E_n, L_{E_0} \dots L_{E_{n-1}} E_n) = \text{Hom}^i(E_n, E_n)$ $\downarrow$ $\text{Hom}(E'_0, E_n) = 0$

Another way: $(E_0, E_1) \quad (L_{E_0} E_1, E_0) \quad (SE_1, E_0)$

Ex. In this case for $X = E_0, E_1$.

- 1) E_0 $\text{Hom}(\dots)$
- 2) E_1

The action on the set of full exceptional collection of length $n+1$ of the central element of B_{n+1}

$$\beta^2 = (\beta_{n, n+1} \beta_{n-1, n} \dots \beta_{2, 3} \beta_{1, 2})^{n+1} \quad \beta = \dots$$

can be extended to a Serre functor S of the category $\mathcal{D}$.

3. Euler Pairing.

- Grothendieck group $K_0(\mathcal{D})$: the quotient of free abelian group generated by $[\mathcal{D}]$ (the set of isomorphism classes of objects of $\mathcal{D}$), with Euler relations:

$$[B] = [A] + [C], \text{ provided with an exact triangle.}$$

$$A \rightarrow B \rightarrow C \rightarrow A[1] \quad \left(\text{or} \quad \begin{array}{ccc} & C & \\ \swarrow & & \searrow \\ A & \xrightarrow{\quad} & B \end{array} \right).$$

By definition we have universal property

$$\begin{array}{ccc} [\mathcal{D}] & \xrightarrow{\varphi} & G \\ \downarrow & \nearrow \exists! \tilde{\varphi} & \\ K_0(\mathcal{D}) & & \end{array}$$

exact functor.
A ~~functor~~ between two triangulated category.

induces a group homomorphism between two.

Grothendieck groups.

Facts 1) $[0] = 0$.

2) $[A[1]] = -[A]$

since $A \rightarrow 0 \rightarrow A[1] \rightarrow$

$A[1]$

If $\mathcal{D}$ is $\mathbb{K}$ -linear. then we can define Euler pairing as.

$$\chi(E, F) := \sum_i (-1)^i \dim_{\mathbb{K}} \text{Hom}^i(E, F).$$

$$\left(\begin{array}{l} \text{has enough injectives in } \mathcal{D} \text{ derived category} \\ = \sum_i (-1)^i \dim \text{Ext}^i(E, F) \\ H^i(\text{Rhom}(E, F)). \end{array} \right).$$

In the case of coherent sheaves on X , we can directly define it as.

For $F^\bullet \in D^b(X)$ $F^\bullet$ be aband. complex of coherent sheaves F^i

$$[F] := \sum (-1)^i [F^i] \in K_0(X).$$

$\rightarrow$ tells us. $K_0^b(\text{coh}) \cong K_0(\text{coh})$.

don't
write
down

• Prop If $\mathcal{D}$ admits a full exceptional collection $(E_0, \dots, E_n)$, then $K_0(\mathcal{D})$ is freely generated by $[E_0], \dots, [E_n]$.

Proof: $\varphi: \mathbb{Z}^n \longrightarrow K_0(\mathcal{D}), \quad e_i \longmapsto [E_i]$.

since $\{E_i\}$ generates $\mathcal{D}$, so every element comes from shifts and mapping cones.

if we have a shift X

$$\begin{array}{c} X \\ \swarrow \quad \searrow \\ E_i[k_i] \longrightarrow E_j[k_j] \end{array}$$

X can be represented by a linear sum of $[E_i]$.

thus φ is surjective.

on the other hand, define $\psi: K_0(\mathcal{D}) \longrightarrow \mathbb{Z}^n$ by

$$[A] \longmapsto (\chi(E_0, A), \dots, \chi(E_n, A)). \quad \text{Consider } \psi \circ \varphi: \mathbb{Z}^n \rightarrow \mathbb{Z}^n$$

then by definition of Euler pairing we have

$$\psi \circ \varphi(e_i) = \sum_{j=1}^n \chi(E_j, E_i) e_j \quad \neq$$

$$\chi(E_j, E_i) = \sum_{k \in \mathbb{Z}} \dim(-1)^k \dim \operatorname{Hom}^k(E_j, E_i).$$

$$\text{when } j > i, \operatorname{Hom}^k(E_j, E_i) = 0. \quad j = i \text{ and } k \neq 0 \quad \operatorname{Hom}^k(E_j, E_i) = \begin{cases} \mathbb{K} & k=0 \\ 0 & k \neq 0 \end{cases}$$

$$\Rightarrow \psi \circ \varphi \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix} = \begin{pmatrix} \\ \\ \end{pmatrix} \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix}$$

$$\psi \circ \varphi(e_1, \dots, e_n) = (e_1, \dots, e_n) \begin{pmatrix} 1 & & & \\ & \ddots & & \\ & & 1 & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix} \begin{matrix} \chi(E_j, E_i) \\ * \\ \vdots \\ 1 \end{matrix}_{(j,i)}$$

thus $\psi \circ \varphi$ invertible $\Rightarrow \varphi$ is injective.

$\Rightarrow (K_0(\mathcal{D}), \chi(-, -))$ is an exceptional unimodular Mukai lattice.

• Relation with mutation.

Prop. $E \in \text{Ob}(\mathcal{D})$ be an exceptional object, then for any object X in $\mathcal{D}$.

we have. $[L_E X] = [X] - \chi(E, X) \cdot [E]$

$$[R_E X] = [X] - \chi(X, E) [E].$$

Proof. ~~$[L_E X]$~~ Since we have exact triangle.

$$L_E X[-1] \longrightarrow \text{Hom}^i(E, X) \otimes E \longrightarrow X \longrightarrow L_E X.$$

$$[L_E X] = [X] - [\text{Hom}^i(E, X) \otimes E].$$

↓ by definition

$$= [X] - [(\bigoplus_i \text{Hom}^i(E, X)) \otimes E]$$

↓ definition

$$= [X] - [\bigoplus_i E[-i] \otimes \dim_{\mathbb{K}} \text{Hom}^i(E, X)]$$

↓ by shift and direct sum.

$$= [X] - \left\{ \underbrace{\sum_i (-1)^i \dim_{\mathbb{K}} \text{Hom}^i(E, X)}_{\chi(E, X)} \right\} \cdot [E].$$

#

$(-1)^i$ (to the power of i)

Prop. $\chi(A, B) = \chi(B, SA)$ for A, B in $\text{Ob}(\mathcal{D})$.